

Error Metrics and Combined Inference

Part III ■ Multiple Testing and Simulation-Based Inference

Richard Luger

Workshop ■ University of Melbourne

Université Laval, Canada

Why look beyond FWER?

FWER: a recap and its cost

What we have been targeting. The **familywise error rate** is the probability of at least one false rejection among m tests:

$$\text{FWER} = P(V \geq 1),$$

where V is the number of true null hypotheses rejected.

The cost. FWER control treats *even one* false rejection as unacceptable. This is appropriate in *confirmatory* settings where a single false positive has severe consequences.

In *exploratory* settings most nulls may be false. Requiring $V = 0$ with high probability then wastes power on errors we are unlikely to make.

FWER is a conservative guarantee designed for the worst case.
When many nulls are false, it over-protects against errors we are unlikely to make.

A spectrum of error criteria

Criterion	Target quantity	Guarantee
FWER	$P(V \geq 1)$	Probability of <i>any</i> false rejection $\leq \alpha$
k -FWER	$P(V \geq k)$	Probability of k or more false rejections $\leq \alpha$
FDP control	$P(\text{FDP} > \gamma)$	Probability FDP exceeds γ is $\leq \alpha$
FDR	$E(\text{FDP})$	Expected proportion of false rejections $\leq \alpha$

R = total rejections, V = false rejections, $\text{FDP} = V/R$ (with $\text{FDP} = 0$ when $R = 0$).

Moving down generally relaxes the guarantee; the right choice depends on the **cost of a false positive**.

Error-metric choice is a scientific judgment, not a statistical default.
FWER when false positives are catastrophic; FDR when they are tolerable.

k-FWER and FDP: procedures based on marginal p -values

k -FWER: tolerating up to $k - 1$ false rejections

Definition. The k -FWER (also written gFWER_k) is the probability of making k or more false rejections. FWER is the special case $k = 1$. Controlling k -FWER at level α allows up to $k - 1$ false rejections with high probability.

One-step procedure (Lehmann & Romano, 2005). Reject $H_{(j)}$ if

$$p_{(j)} \leq \frac{k\alpha}{m}.$$

Controls k -FWER under *any* dependence. For $k = 1$ this is **Bonferroni**. The general case applies the Bonferroni union bound to the event $\{V \geq k\}$ rather than $\{V \geq 1\}$, replacing α by $k\alpha$.

Step-down improvement (Lehmann & Romano, 2005). Find the largest φ such that $p_{(j)} \leq \alpha_j$ for all $j = 1, \dots, \varphi$, where

$$\alpha_j = \frac{k\alpha}{m - j + k}, \quad j = 1, \dots, m.$$

Reject $H_{(1)}, \dots, H_{(\varphi)}$. Valid under arbitrary dependence; uniformly more powerful than the one-step rule. For $k = 1$ the thresholds $\alpha_j = \alpha / (m - j + 1)$ reduce to **Holm**.

FDP: controlling the proportion of false rejections

Definition.

$$\text{FDP} = \frac{V}{R}, \quad \text{FDP} = 0 \text{ if } R = 0.$$

FDP control (Lehmann & Romano, 2005) requires

$$P(\text{FDP} > \gamma) \leq \alpha.$$

Step-down procedure. Use the critical values

$$\alpha_j = \frac{(\lfloor \gamma j \rfloor + 1)\alpha}{m + \lfloor \gamma j \rfloor + 1 - j}, \quad j = 1, \dots, m.$$

Find the largest φ such that $p_{(j)} \leq \alpha_j$ for $j = 1, \dots, \varphi$, and reject $H_{(1)}, \dots, H_{(\varphi)}$.

If $R = j$, then $\text{FDP} > \gamma$ requires at least $\lfloor \gamma j \rfloor + 1$ false rejections. The critical values control this event by a Bonferroni argument, under arbitrary dependence.

Setting $\gamma = 0$ gives Holm's FWER procedure; larger γ allows more false rejections as a fraction of all rejections.

**FDP control bounds the probability that the error proportion exceeds γ .
FDR control relaxes this by targeting only the expected error proportion.**

***k*-FWER and FDP: procedures based on the joint distribution**

Going beyond marginal p -values

Lehmann–Romano procedures use marginal p -values and Bonferroni bounds. This is valid under arbitrary dependence, but conservative because it ignores correlation among tests.

The same idea as max- t . Max- t improves on Bonferroni by replacing a marginal bound with the joint null distribution. The same idea applies to k -FWER and FDP control.

What changes.

- The *target* stays the same: control k -FWER or FDP.
- The *calibration* changes: replace the Bonferroni bound by a simulated joint null distribution.
- For k -FWER, use the k -max statistic: the k -th largest statistic under the joint null.
- With an exact pivot or valid permutation scheme, finite-sample validity is inherited.

**Marginal procedures bound the error probability.
Joint procedures estimate the relevant null distribution directly.**

Simulation-based k -FWER

Single-step k -FWER. For each simulated sample, compute

$$T_{k\text{-max}} = k\text{-th largest value among } \{T_1, \dots, T_m\}.$$

The adjusted p -value for hypothesis H_j is

$$\tilde{p}_{SS,j}^k = P\left(T_{k\text{-max}} \geq T_j^{\text{obs}} \mid H_0\right).$$

Thus every hypothesis is compared with the same joint-null reference distribution. This is the simulation analogue of a single-step k -FWER adjustment.

Step-down k -FWER. First sort the observed statistics: $T_{(1)}^{\text{obs}} \geq \dots \geq T_{(m)}^{\text{obs}}$. The first k hypotheses are compared with the full-sample k -max reference. For later hypotheses, the reference statistic is based on the largest simulated statistic among the hypotheses still under consideration:

$$\max_{\ell \geq j} T_{(\ell)}^{(b)}, \quad j > k,$$

with the usual monotonicity correction imposed on the adjusted p -values.

Single-step k -FWER: every test uses the same full-sample k -max reference.

Step-down k -FWER: after the first k tests, the reference is updated over progressively smaller remaining sets.

Simulation-based FDP control

Sequential k -FWER. For $k = 1, 2, \dots$, apply the simulation-based k -FWER procedure using the same simulated draws. Let

$$R_k = \#\{j : H_j \text{ is rejected by the } k\text{-FWER procedure}\}.$$

Stopping rule. Continue while $k \leq \gamma(R_k + 1)$. Stop when $k > \gamma(R_k + 1)$ and set $k^* = k - 1$. The FDP-adjusted values are the k^* -FWER adjusted values.

Why it works. For the selected k^* ,

$$k^* \leq \gamma(R_{k^*} + 1) \quad \Rightarrow \quad k^* - 1 \leq \gamma R_{k^*}.$$

The k^* -FWER guarantee gives $V \leq k^* - 1$ with probability at least $1 - \alpha$. Therefore,

$$\text{FDP} = \frac{V}{R_{k^*}} \leq \frac{k^* - 1}{R_{k^*}} \leq \gamma.$$

FDP control selects the largest admissible k .

The same simulated draws are reused for coherence across k .

False Discovery Rate

The FDR: the BH algorithm

Definition (Benjamini & Hochberg, 1995). The false discovery rate is

$$\text{FDR} = E[\text{FDP}].$$

Controlling FDR at level α means controlling the expected proportion of false discoveries among the rejected hypotheses.

The BH algorithm. Given ordered p -values $p_{(1)} \leq \dots \leq p_{(m)}$, define

$$j_{\max} = \max \left\{ j : p_{(j)} \leq \frac{j\alpha}{m} \right\}.$$

Reject $H_{(1)}, \dots, H_{(j_{\max})}$. If no such j exists, reject nothing.

What BH delivers. Under independence, $\text{FDR} \leq \pi_0 \alpha \leq \alpha$ where $\pi_0 = m_0/m$ is the proportion of true nulls. When many nulls are false, $\pi_0 \ll 1$ and BH is conservative: the actual FDR is well below α .

Under arbitrary dependence, Benjamini & Yekutieli (2001) replace $j\alpha/m$ by $j\alpha/(mc_m)$ where $c_m = \sum_{j=1}^m 1/j \approx \log m + 0.577$. The harmonic penalty accounts for the worst-case dependence structure; for $m = 50$, $c_m \approx 4.5$, so BY requires p -values roughly 4.5 times smaller than BH to be rejected.

BH interpolates between Bonferroni (rank 1) and the unadjusted level (rank m), controlling the expected proportion of false rejections.

The BH threshold: a geometric view

Sort the m p -values and plot them against rank.

Draw the BH line $y = j\alpha/m$.

Rule: find the last rank where the ordered p -value lies on or below the BH line. Reject all hypotheses up to that rank.

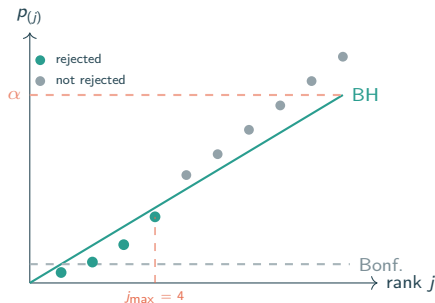
The BH line interpolates. At rank $j = 1$ the threshold is α/m : Bonferroni.

At rank $j = m$ the threshold is α : the unadjusted level.

For all ranks in between, BH is strictly more permissive than Bonferroni:

$$\frac{j\alpha}{m} > \frac{\alpha}{m}, \quad j \geq 2.$$

The line trades off stringency against rank: the more hypotheses rejected, the more lenient the threshold for the marginal rejection.



Applied illustration: Clements (2024)

Clements (2024): the model and the MT problem

Setting. U.S. Survey of Professional Forecasters (SPF), up to $m = 50$ respondents. For each respondent i , estimate one of two regressions:

Forecast efficiency (Mincer–Zarnowitz):

$$y_{t+h} - \hat{y}_{i,t+h|t} = \mu_i + \beta_i \hat{y}_{i,t+h|t} + e_{it}.$$

$H_0 : \beta_i = 0$: forecast error unrelated to the forecast level.

Over-reaction (revision regression):

$$y_{t+h} - \hat{y}_{i,t+h|t} = \mu_i + \beta_i (\hat{y}_{i,t+h|t} - \hat{y}_{i,t+h|t-1}) + e_{it}.$$

$H_0 : \beta_i = 0$: forecast error unrelated to the revision. Negative $\hat{\beta}_i$ indicates over-reaction.

The MT problem. With $m \approx 50$ separate tests, testing each at 5% makes false rejections likely by chance. Forecast errors share the realised y_{t+h} and respondents use common public information, so dependence across p -values is plausible. BH is justified under independence-type conditions; BY is safer under arbitrary dependence.

Results: forecast efficiency (Table 3, Panel A)

Proportion of respondents for whom H_0 is rejected at 5%, by correction method. **Output growth**, $h = 0$ vs. $h = 4$.

Correction	$h = 0$	$h = 4$
Liberal (no correction)	0.42	0.94
BH (FDR, indep.-type)	0.32	0.94
BY (FDR, any dep.)	0.20	0.82
FDP ($\gamma = 0.05$)	0.14	0.92
Bonferroni	0.14	0.72
Holm	0.14	0.74

Two patterns.

- $h = 0$: BH stays closer to liberal testing ($0.42 \rightarrow 0.32$); BY and FWER methods reduce rejections to 0.14–0.20.
- $h = 4$: MT matters less. Even Bonferroni leaves many rejections ($0.94 \rightarrow 0.72$), consistent with many false nulls.

Short horizon: widespread inefficiency becomes a minority finding once corrected.

Long horizon: many rejections survive even conservative MT adjustment.

Choosing your error criterion

FWER. “Every forecaster I flag must really be inefficient. I accept low power.”

⇒ Bonferroni / Holm. $0.42 \rightarrow 0.14$.

FDR. “I expect heterogeneity. Most rejections should be real; a few false calls are acceptable.”

⇒ BY is the safer FDR choice under dependence. $0.42 \rightarrow 0.20$.

Dependence warning. SPF respondents share realised outcomes and common public information, so the independence-style justification for BH is questionable. BY is valid under arbitrary dependence.

Bottom line (Clements, 2024). Some apparent violations of rationality are sensitive to multiple-testing adjustment; others are robust. The adjustment is simple and can materially change the conclusions.

**Pick the criterion before seeing the data.
Use BH only when independence-type conditions are credible.**

The combination problem

The combination problem

A different setting from Clements. Clements tests one hypothesis per forecaster: autonomous entities, exploratory goal. Here m diagnostics assess the *same* model: the verdict is binary, and a single false rejection rejects the model. FWER is the natural criterion.

The implementation problem. Diagnostics share estimated residuals and are therefore dependent. Bonferroni is valid under any dependence but ignores correlation, over-correcting when tests are positively correlated.

The natural combined statistic. Rejecting the model requires evidence against at least one diagnostic, which maps directly onto

$$p_{\min} = \min_i p_i,$$

the natural companion to FWER. Under H_0 each $p_i \stackrel{a}{\sim} U(0, 1)$, but dependence means $p_{\min}$ has no simple null distribution.

The statistic is simple.

The difficulty is its null distribution under dependence.

Monte Carlo combination tests

Restoring exactness: the MC-test key

The MC test. If the null distribution of a statistic is nuisance-parameter-free and can be simulated, draw $B - 1$ replicates and rank the observed combined statistic $\bar{S}_B$ among all B values. The MC p-value

$$\hat{p}(\bar{S}_B) = \frac{B - \hat{R}(\bar{S}_B) + 1}{B}$$

satisfies $P[\hat{p}(\bar{S}_B) \leq \alpha] = \alpha$ *exactly* when αB is an integer.

The insight of Dufour, Khalaf & Voia (2015). Apply the MC test to the *combined* statistic $\bar{S} = g(p_1, \dots, p_m)$ rather than to an individual statistic.

If the joint null distribution of $(S_1, \dots, S_m)$ is nuisance-parameter-free, the combined statistic inherits the same property, and **exact size control is restored** regardless of the dependence structure among the m diagnostics.

Applying the MC test to the combined criterion converts a conservative bound (Bonferroni) into an exact finite-sample guarantee.

Theorem: exact finite-sample validity

Theorem (Dufour, Khalaf & Voia, 2015)

Let $S_1, \dots, S_m$ be test statistics for $H_{01}, \dots, H_{0m}$ (not necessarily independent). Let $\bar{S} = g(S_1, \dots, S_m)$ be any combination rule from the Tippett / Fisher / weighted family.

If, under H_0 , the joint distribution of $(S_1, \dots, S_m)$ is unique and free of nuisance parameters, and $\bar{S}_1, \dots, \bar{S}_{B-1}, \bar{S}_B$ are exchangeable with $P[\bar{S}_i = \bar{S}_j] = 0$ for $i \neq j$, then

$$P[\hat{p}(\bar{S}_B) \leq \alpha] = \alpha$$

whenever αB is an integer.

Proof sketch. Exchangeability implies all rankings are equally likely: $P[\hat{R}(\bar{S}_B) = k] = 1/B$ for $k = 1, \dots, B$. $\square$

$B = 20$ suffices for $\alpha = 5\%$; $B = 100$ is typical in practice.

The key subtlety: approximate individual p-values are fine

The combination rules use individual p-values p_i as inputs, which may be asymptotic, bootstrap-estimated, or exact.

None of this breaks the theorem. As long as each p_i is computed *the same way* in the observed sample and in every MC replication, exchangeability is preserved and exact size control is retained. The simulation absorbs the individual approximation errors.

Practical consequence. Use cheap asymptotic χ^2 or F p-values for individual diagnostics, combine via Tippett or Fisher, and simulate with $B = 100$ replications. The overall test is exact.

An exact combined test does not require exact individual tests.

**Uniformity of treatment across observed
and simulated samples is all that matters.**

Application: serial correlation in the CLR model

Why the CLR model works. In $y_t = x_t'\beta + u_t$ with $u_t = \sigma\varepsilon_t$ and $\varepsilon_t \stackrel{i.i.d.}{\sim} F_0$ (completely specified), the scaled residual vector $\hat{u}/\hat{\sigma}$ depends only on F_0 and X , not on (β, σ) : any statistic built from it is **jointly pivotal**.

Test battery (Dufour, Khalaf & Voia, 2015):

Label	Individual p_i from	Rule
$AC_{\min}, AC_{\times}$	$\hat{\rho}_j^2$ (asymptotic normal)	Tippett / Fisher
$VR_{\min}, VR_{\times}$	$ VR(j) $ (asymptotic normal)	Tippett / Fisher

$B = 100$ MC replications $\Rightarrow$ exact size at $\alpha = 5\%$.

Power gains. Asymptotic tests are either oversized (Ljung-Box under normal/ χ^2 /uniform errors) or severely undersized (variance-ratio under all error distributions). MC versions have exact size throughout, directly raising power for the previously undersized tests.

The Cauchy combination test

An analytic shortcut: the Cauchy transform

The key observation. If $p_i \sim U(0, 1)$ under H_0 , define

$$V_i = \tan\left\{\left(\frac{1}{2} - p_i\right) \pi\right\}.$$

Then $V_i \sim \text{Cauchy}(0, 1)$ *exactly*. In practice p_i is often asymptotic, so V_i is approximately Cauchy.

The CCT statistic (Liu & Xie, 2020)

$$T = \sum_{i=1}^m \omega_i V_i = \sum_{i=1}^m \omega_i \tan\left\{\left(\frac{1}{2} - p_i\right) \pi\right\}, \quad \omega_i \geq 0, \quad \sum_{i=1}^m \omega_i = 1.$$

Here ω_i is the weight assigned to test i ; equal weights $\omega_i = 1/m$ are the default when all diagnostics are equally important.

A single very small p_i creates a large positive V_i , so the Cauchy sum is sensitive to sparse signals. The p-value is

$$\text{p-value} \approx \frac{1}{2} - \frac{\arctan(t_0)}{\pi}.$$

One arctangent. No simulation. Validity is asymptotic.

Why the log transform? Why the Cauchy transform?

Fisher: the log transform. Under H_0 , each $p_i \sim U(0, 1)$, so $-2 \log p_i \sim \chi^2(2)$. Independent $\chi^2(2)$ variables sum to $\chi^2(2m)$:

$$-2 \sum_{i=1}^m \log p_i \sim \chi^2(2m) \quad (\text{under independence}).$$

Under dependence the $\chi^2(2m)$ reference fails: simulate instead.

Tippett: focus on the minimum. Under independence, $p_{\min} \sim \text{Beta}(1, m)$ with exact threshold $\alpha_0 = 1 - (1 - \alpha)^{1/m}$. Under dependence the $\text{Beta}(1, m)$ reference fails: simulate instead.

CCT: the Cauchy transform (Liu & Xie, 2020). If $p_i \sim U(0, 1)$ then $V_i = \tan\{(\frac{1}{2} - p_i)\pi\} \sim \text{Cauchy}(0, 1)$. A weighted sum of Cauchy variables is also Cauchy (**stability property**). Under regularity conditions on the dependence structure, the tail of T remains asymptotically Cauchy:

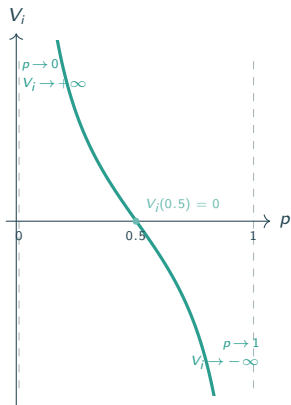
$$\frac{1}{2} - \frac{\arctan(T)}{\pi}$$

is asymptotically valid without simulation and without knowing Σ .

**Fisher and Tippett need simulation when tests are dependent.
The CCT avoids simulation, but validity is asymptotic
and requires regularity on the dependence structure.**

The Cauchy transform: what it does to p-values

The map $p \mapsto \tan\left\{\left(\frac{1}{2} - p\right)\pi\right\}$:



Why this matters for power.

A single very small p_i maps to a very large positive V_i , which can alone push T into the rejection region.

This makes the CCT **especially powerful against sparse alternatives**: when only one or two tests fire, those alone can move T enough to reject.

Contrast with Fisher. $-2 \ln p_i$ also sends small p -values to large values, but the $\chi^2(2m)$ reference breaks down under dependence. The Cauchy tail is asymptotically robust to dependence under regularity; the χ^2 reference is not.

Power optimality (Liu & Xie, 2020)

Sparse alternatives. Let $|S| = m^\gamma$ signals with $0 < \gamma < \frac{1}{2}$, each of magnitude $|\mu_i| = \sqrt{2r \log m}$. If $r > (1 - \sqrt{\gamma})^2$, then

$$\lim_{m \rightarrow \infty} P[\text{CCT rejects}] = 1, \quad \forall \alpha > 0.$$

The CCT attains the **optimal detection boundary** of Donoho & Jin (2004): no test can do better in the sparse regime.

This optimality survives dependence. Under bivariate normality for all pairs (X_i, X_j) , a regularity condition not a universal guarantee, the tail of T is asymptotically Cauchy regardless of Σ :

$$\lim_{t \rightarrow +\infty} \frac{P(T > t)}{P(W_0 > t)} = 1, \quad W_0 \sim \text{Cauchy}(0, 1).$$

The null shape is stable across dependence structures, so no simulation is needed to calibrate it.

Consequence (Long et al., 2023). $\beta_{\text{CCT}} \geq \beta_{\text{MinP}} + o(1)$ as $m \rightarrow \infty$: simulating the min- p null buys no additional power against sparse signals. One arctangent suffices.

Against sparse alternatives, CCT is optimally powerful, dependence-robust, and simulation-free. Simulation buys nothing here.

Combining dependent tests: two frameworks

	Dufour et al. (2015)	CCT (Liu & Xie, 2020)
Size validity	Exact finite-sample	Asymptotic
Validity condition	Joint pivotality	Regularity on dependence
Calibration	Monte Carlo ranking	Analytic Cauchy tail
Simulation needed	Yes	No
Combination rule	User-chosen	Cauchy-weighted sum
Sparse alternatives	Tippett / min- p	Very strong
Dense alternatives	Fisher	Usually weaker

**Dufour et al. give exact finite-sample control under joint pivotality.
CCT gives an analytic shortcut but validity is asymptotic and requires regularity on the dependence structure.**

Demo 3: setup (1)

The null model. Classical linear regression with $T = 60$, $k = 3$ regressors, and $m = 5$ autocorrelation lags tested:

$$y_t = x_t' \beta + u_t, \quad u_t = \sigma \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2), \quad \sigma > 0 \text{ unknown.}$$

The scaled residual vector $\hat{u}/\hat{\sigma}$ is jointly pivotal: $B = 100$ MC draws give exact size at $\alpha = 5\%$.

Three combination rules applied to five p-values.

Rule	Statistic	Rejects when
Tippett	$-\log p_{\min}$	minimum p-value is small
Fisher	$-2 \sum_{j=1}^5 \log p_j$	aggregate evidence is large
CCT	$\frac{1}{5} \sum_{j=1}^5 \tan\{(\frac{1}{2} - p_j)\pi\}$	Cauchy sum is large

Same data, same five tests, three combination rules.
The question: does the choice of rule matter for power?

Demo 3: setup (2)

Individual test statistics. For each lag $j = 1, \dots, 5$:

$$\hat{\rho}_j = \frac{\sum_{t=j+1}^T z_t z_{t-j}}{\sum_{t=1}^T z_t^2}, \quad p_j = 2(1 - \Phi(\sqrt{T}|\hat{\rho}_j|)).$$

Each $p_j \stackrel{a}{\sim} U(0, 1)$ under H_0 . Uniform treatment across observed and simulated samples restores exact size.

Two alternatives: errors ε_t follow an MA process.

Sparse — MA(1):

$$\varepsilon_t = \frac{\eta_t + \theta \eta_{t-1}}{\sqrt{1 + \theta^2}}, \quad \eta_t \stackrel{i.i.d.}{\sim} N(0, 1), \quad \theta \in \{0.25, 0.35, 0.45, 0.55\}.$$

$\rho(1) = \theta/(1 + \theta^2)$; $\rho(2) = \dots = \rho(5) = 0$ exactly.

Dense — MA(5):

$$\varepsilon_t = \frac{\eta_t + \theta \sum_{l=1}^5 \eta_{t-l}}{\sqrt{1 + 5\theta^2}}, \quad \theta \in \{0.10, 0.15, 0.20, 0.25\}.$$

All five coefficients equal θ ; autocorrelations decline from $\rho(1)$ to $\rho(5)$ as signal spreads across lags.

MA(1) is pure sparse. MA(5) is pure dense.
The demo sweeps θ to show how power grows with signal strength.